INDIRA GANDHI NATIONAL OPEN UNIVERSITY

School of Computer and Information Sciences

Master of Science (Data Science and Analytics) — MSCDSA

JULY 2026 / SEMESTER I

MCSL-065

Data Science Lab

Lab assignment and practical write-up

Student particulars	Details
Name	
Enrolment number	
Programme	MSCDSA
Regional Centre	38 — Delhi 3 (Naraina)
Study Centre code / name	_____
Session	July 2026
Assignment number	MSCDSA(I)/L-065/Lab_Assignment/2026
Due date in the supplied paper	31 October 2026
Submission date	_____

Assessment: assignment questions 40 marks + lab record 40 marks + viva voce 20 marks = 100 marks. Assignment weightage: 30%.

Review copy — native Excel, Tableau and Power BI screenshots still required

Student signature: _____

The Regional Centre is recorded separately from the Study Centre. Fill the Study Centre and sign/date the report before submission.

Record map and scope

Section / question	Marks	Evidence included
Q1 — Excel	6	Workbook, four native charts, formulas, statistics and tests; actual Excel ToolPak capture pending.
Q2 — Tableau	7	Data, exact worksheet/dashboard instructions and expected values; native workbook/screenshots pending.
Q3 — Power BI	7	Import queries, all 16 subject measures, display measures and report design; Desktop validation/screenshots pending.
Q4 — Python	10	Complete executed code, captured output, four plots and checked results.
Q5 — R	10	Complete executed base R code, captured output, four plots and checked results.

The provided Excel workbook was generated with native Excel chart objects and formulas. It was recalculated and rendered in LibreOffice Calc: all 122 formula values matched. These renderings are identified accurately and are not presented as Microsoft Excel screenshots.

The Tableau and Power BI sections are build-ready instructions and source drafts. They have not been executed in their desktop applications in this cloud environment. Their remaining captures must be inserted into this editable report before submission.

The supplied question paper explicitly requires screenshots with coding and documentation. This review copy therefore distinguishes calculated results from native application evidence. The separate Completion_Checklist.md lists the final steps.

Statistical conventions

Variance and standard deviation use the sample definition (denominator $n - 1$). A mode is reported only if a value occurs more than once; all tied modal values would be retained. Statistical tests use $\alpha = 0.05$. Charts use the same subject colours across outputs.

Dataset and initial checks

ID	Name	Physics	Chem.	Maths	English	Total	%
S01	Rahul	78	72	85	80	315	78.75
S02	Ananya	88	91	90	86	355	88.75
S03	Amit	65	69	70	72	276	69.00
S04	Neha	92	89	94	90	365	91.25
S05	Karan	55	60	58	62	235	58.75
S06	Priya	81	85	88	84	338	84.50
S07	Rohit	70	68	75	73	286	71.50
S08	Simran	90	92	89	91	362	90.50
S09	Arjun	60	65	63	67	255	63.75
S10	Pooja	85	88	90	87	350	87.50

Source: Students' Exam Scores in the supplied 2026 MCSL-065 assignment paper. There are 10 students, 4 subjects and no missing values. Student_ID and Student_Name are text; subject marks are whole numbers from 0 to 100. The CSV retains all names and scores exactly as printed.

Derived columns

```
Total = Physics + Chemistry + Maths + English
Percentage = (Total / 400) * 100
```

Neha (S04) is the topper with 365/400 (91.25%). Karan (S05) has the lowest total, 235/400 (58.75%). The class mean total is 313.70 and mean percentage is 78.425%.

Student IDs refer to different individuals. A line joining their marks shows differences across students in ID order; it does not show improvement or decline over time.

Q1 | Excel workflow

1(a–d). Histogram and charts

Open Students_Exam_Scores.xlsx. Scores contains the input data and formula-based totals/percentages. Dashboard contains the average bar chart, Maths line chart and contribution pie. Histogram contains grouped frequencies and a zero-gap column chart representing the histogram.

Task	Excel steps / formula	Expected check
Maths histogram	Use Scores!E2:E11. Group marks in 10-mark intervals: 50–59, 60–69, 70–79, 80–89, 90–100.	Frequencies 1, 1, 2, 3, 3.
Maths line chart	Select Student_ID and Maths; Insert → Line with Markers. Label axes Student ID and Maths marks / 100.	Lowest S05 = 58; highest S04 = 94.
Subject average bar chart	=AVERAGE(C2:C11), and similarly for D, E and F. Plot the four subject means.	Maths highest 80.20; Physics lowest 76.40.
Contribution pie	Use the four means as the pie values. Share = subject mean / sum of four means. Display percentage labels.	Maths share = 80.2/313.7 = 25.57%.

1(e). Run the Analysis ToolPak in Microsoft Excel

Enable File → Options → Add-ins → Manage Excel Add-ins → Go → Analysis ToolPak. Then use Data → Data Analysis. Choose labels in first row whenever the ranges below include row 1.

Descriptive Statistics: Scores!C1:F11; group by columns; select Summary statistics.
t-Test: Two-Sample Assuming Equal Variances: Physics C1:C11 versus Chemistry D1:D11; hypothesized difference 0; α = 0.05.
F-Test Two-Sample for Variances: Maths E1:E11 versus English F1:F11; α = 0.05.
Correlation: use the marks columns, or calculate CORREL(E2:E11,C2:C11) and CORREL(E2:E11,D2:D11).

The workbook has formula equivalents for checking. Capture the actual Excel charts and ToolPak result sheets after running the tools; the ToolPak was not executed here.

Q1 | Summary and inference

Subject	Mean	Median	Mode	Min	Max	Range	Variance	SD
Physics	76.40	79.50	No mode	55	92	37	173.1556	13.1589
Chemistry	77.90	78.50	No mode	60	92	32	149.4333	12.2243
Maths	80.20	86.50	90	58	94	36	162.6222	12.7523
English	79.20	82.00	No mode	62	91	29	102.4000	10.1193

Physics, Chemistry and English have no repeated score and hence no mode under the stated convention. Maths has mode 90, occurring twice. Excel’s MODE.SNGL gives #N/A when no value repeats; the workbook displays “No mode” using IFERROR.

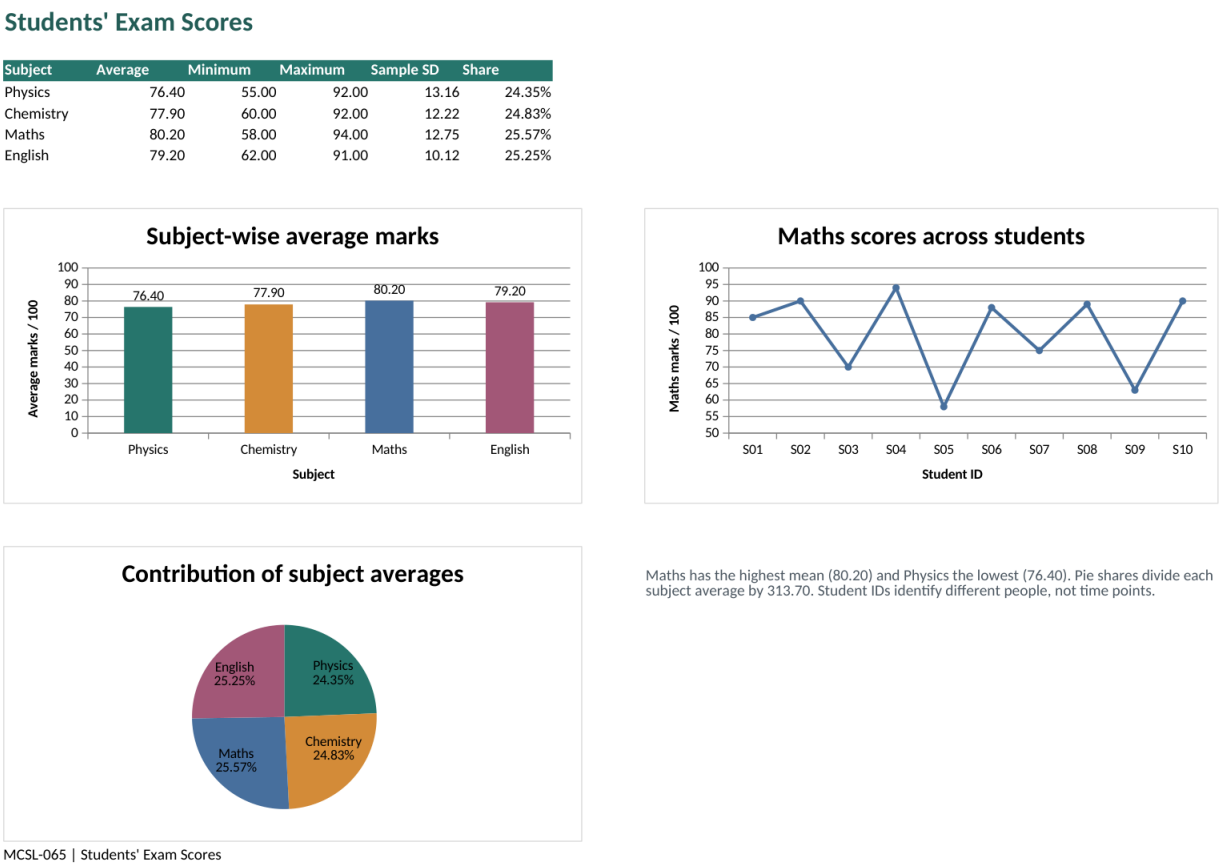
Analysis	Calculated result	Conclusion at α = 0.05
Two-sample equal-variance t-test	t = −0.264099; df = 18; two-sided p = 0.794704; critical |t| = 2.100922.	Fail to reject equal population means under the requested model.
F-test: Maths / English	F = 1.588108; df1 = df2 = 9; upper-tail p = 0.250805; two-sided p = 0.501609.	Fail to reject equal variances in a two-sided test.
Pearson: Maths–Physics	r = 0.982085	Strong positive linear association in these 10 records.
Pearson: Maths–Chemistry	r = 0.921742	Strong positive linear association in these 10 records.

Interpretation and limitations

A large p-value means the observed test statistic is not unusual under the null model; it is not the probability that the null hypothesis is true. Failing to reject does not establish equality. Correlation describes association and does not establish causation.

The two subjects were measured for the same students, so a paired comparison is more appropriate for a substantive within-student question. The independent two-sample test is included because the assignment explicitly requests it. The F-test also assumes normal independent observations and is sensitive to violations. Findings should be treated as a small instructional example.

Q1 | Workbook chart output



Students' Exam Scores

Subject	Average	Minimum	Maximum	Sample SD	Share
Physics	76.40	55.00	92.00	13.16	24.35%
Chemistry	77.90	60.00	92.00	12.22	24.83%
Maths	80.20	58.00	94.00	12.75	25.57%
English	79.20	62.00	91.00	10.12	25.25%

Figure 1. Actual rendering of the generated Excel workbook in LibreOffice Calc. This is workbook-content evidence, not a Microsoft Excel interface screenshot.

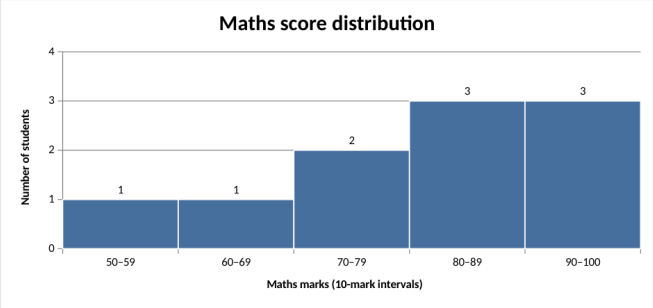
The averages are close: only 3.80 marks separate the highest mean (Maths, 80.20) from the lowest (Physics, 76.40). Pie shares are 24.35%, 24.83%, 25.57% and 25.25%, respectively. Subject shares describe contributions of the four means; they are not student pass rates.

Native Excel evidence to add: capture the Dashboard worksheet with its three charts and the Histogram sheet, followed by the ToolPak output.

Q1 | Histogram and test output

Maths histogram

Interval	Upper bound	Frequency
50–59	59	1
60–69	69	1
70–79	79	2
80–89	89	3
90–100	100	3



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Figure 2. Workbook histogram rendering in LibreOffice Calc; frequencies total 10.

Hypothesis tests and correlations

Two-sample t-test	Physics	Chemistry
Mean	76.40	77.90
Variance	173.16	149.43
Observations	10.00	10.00
Hypothesized mean difference	0	
Pooled variance	161.294444	
Degrees of freedom	18	
t statistic	-0.264099	
Two-sided p-value	0.794704	
Two-sided critical |t| (5%)	2.100922	

At alpha = 0.05, the requested independent two-sample equal-variance t-test fails to reject equal means. The two-sided F-test fails to reject equal variances. Neither result proves equality. Scores come from the same students, so independence is a limitation: a paired comparison is more appropriate for a substantive within-student question. Normality is also an assumption with only 10 records.

F-test	Maths	English
Sample variance	162.622222	102.400000
F statistic (Maths / English)	1.588108	
Degrees of freedom (both)	9	
Upper-tail reference p	0.250805	
Two-sided p-value	0.501609	
Two-sided upper critical F	4.025994	
Pearson r: Maths–Physics	0.982085	
Pearson r: Maths–Chemistry	0.921742	

Formula-based outputs checked against SciPy. Run the actual Analysis ToolPak in Excel for the required native evidence. Its F-test may label an upper-tail probability as P(F<=f) one-tail; use the stated tail and compare a two-sided p-value when testing for any variance difference.

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Figure 3. Workbook formula-based test output, rendered in LibreOffice Calc. The two-sided F-test p-value is 0.501609; the upper-tail reference probability is 0.250805.

Q2 | Tableau import and worksheets

Tool: Tableau Desktop or Tableau Public desktop. Native execution and screenshots are pending. (7 marks)

Import and verify

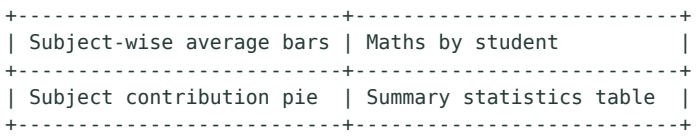
Connect to students_scores.csv as a Text File. Confirm ten records. Student_ID and Student_Name must be strings; Physics, Chemistry, Maths and English must be whole numbers. Select the four subject columns and choose Pivot. Rename Pivot Field Names to Subject and Pivot Field Values to Score. The pivoted data has 40 records, preserving all 10 students × 4 subjects.

Worksheet	Construction	Checks
Subject-wise average	Subject → Columns; AVG(Score) → Rows; Bar marks; Subject → Colour; show value labels.	Averages: Physics 76.40, Chemistry 77.90, Maths 80.20, English 79.20.
Maths across students	Filter Subject = Maths on this sheet only; Student_ID → Columns; AVG(Score) → Rows; Line marks with markers; sort ID ascending.	S04 = 94, S05 = 58, S10 = 90.
Subject contribution	Use all subjects. Pie marks; Subject → Colour; AVG(Score) → Angle and Label; Percent of Total on label, computed across Subject.	Display all four percentage labels; total 100%.
Summary statistics	Subject → Rows; Measure Names → Columns; Measure Values → Text; include AVG(Score), MIN(Score), MAX(Score), STDEV(Score).	STDEV uses the sample definition. Maths SD = 12.75.

Set meaningful chart titles and axes. Use marks out of 100, and show averages/standard deviations to two decimal places. Do not apply the Maths-only filter to all worksheets.

Q2 | Dashboard and validation

Create one dashboard containing the four required worksheets. A practical arrangement is a 1200 × 800 dashboard with average bars and the Maths line on the top row, and the contribution pie and statistics table below. Use consistent colours and retain readable labels.



This is a layout plan, not a screenshot from Tableau.

Subject	Average	Minimum	Maximum	Sample SD	Share
Physics	76.40	55	92	13.16	24.35%
Chemistry	77.90	60	92	12.22	24.83%
Maths	80.20	58	94	12.75	25.57%
English	79.20	62	91	10.12	25.25%

Save the workbook locally, preferably as a packaged workbook with its data. Capture the data source types/pivot and the dashboard with all four views. If a chart’s percent labels do not add to approximately 100%, check the aggregation and the table calculation’s addressing.

Files supplied: students_scores.csv, tableau/scores_long.csv and tableau/BUILD_GUIDE.md. The long-form CSV is a check of the intended pivot. A native Tableau workbook and application screenshots remain to be produced.

Q3 | Power BI import and model

Tool: Power BI Desktop on Windows. Model/measure sources are drafts pending Desktop validation. (7 marks)

Use Get Data → Text/CSV → students_scores.csv → Transform Data. Name the query StudentsScores. Verify Student_ID/Student_Name as Text and each subject as Whole Number. The source has 10 rows. Keep the original subject names; no renaming is needed.

For a reproducible import, define a Text parameter CsvPath with the CSV filename and paste powerbi/StudentsScores.m into Advanced Editor. Create the Subjects query using Subjects.m. It contains four labels and a SortOrder; sort Subject by SortOrder. It is a disconnected selector table and requires no relationship to StudentsScores.

All requested subject measures

Create each definition below as a separate New measure. Repeat for all four subjects, as listed on the following page. These are measures, evaluated in filter context; do not create them as calculated columns.

```
Physics Average = AVERAGE(StudentsScores[Physics])
Physics Minimum = MIN(StudentsScores[Physics])
Physics Maximum = MAX(StudentsScores[Physics])
Physics Std Dev = STDEV.S(StudentsScores[Physics])
```

STDEV.S is used so Power BI agrees with Excel STDEV.S, pandas std(ddof=1), and R sd(). The disconnected Subject selector makes the bar, pie and summary table convenient without duplicating the source rows.

The supplied source design does not contain credentials or require public service publication. Save a local PBIX/PBIP after creating and validating the report in Desktop. No native binary is supplied or claimed to have been tested.

Q3 | Complete subject measures

Sixteen required measures: average, minimum, maximum and sample standard deviation for each subject.

```
// Q3: create these as separate New Measure definitions in Power BI Desktop.
// Draft until checked in Desktop. Standard deviations use the sample definition.

Physics Average = AVERAGE(StudentsScores[Physics])

Physics Minimum = MIN(StudentsScores[Physics])

Physics Maximum = MAX(StudentsScores[Physics])

Physics Std Dev = STDEV.S(StudentsScores[Physics])

Chemistry Average = AVERAGE(StudentsScores[Chemistry])

Chemistry Minimum = MIN(StudentsScores[Chemistry])

Chemistry Maximum = MAX(StudentsScores[Chemistry])

Chemistry Std Dev = STDEV.S(StudentsScores[Chemistry])

Maths Average = AVERAGE(StudentsScores[Maths])

Maths Minimum = MIN(StudentsScores[Maths])

Maths Maximum = MAX(StudentsScores[Maths])

Maths Std Dev = STDEV.S(StudentsScores[Maths])

English Average = AVERAGE(StudentsScores[English])

English Minimum = MIN(StudentsScores[English])

English Maximum = MAX(StudentsScores[English])

English Std Dev = STDEV.S(StudentsScores[English])
```

Q3 | Display measures and report page

```
Subject Average =
SWITCH(SELECTEDVALUE(Subjects[Subject]),
    "Physics", [Physics Average],
    "Chemistry", [Chemistry Average],
    "Maths", [Maths Average],
    "English", [English Average], BLANK())

Subject Contribution % =
DIVIDE([Subject Average], SUMX(ALL(Subjects), [Subject Average]))
```

Subject Minimum, Subject Maximum and Subject Std Dev use the same SWITCH pattern, selecting the matching subject measure. Their complete definitions are in powerbi/measures.dax. Format averages and standard deviations to two decimals and contribution as a percentage.

Visual	Fields	Validation
Average bar chart	Subjects[Subject] + Subject Average	Maths highest 80.20; Physics lowest 76.40.
Maths line	StudentsScores[Student_ID] + Maths Average; sort ascending; markers on	Ten student points. IDs are categories.
Contribution pie	Legend: Subject; Values: Subject Average; enable category and percent-of-total labels	Maths 25.57%; English 25.25%.
Statistics table	Subject + Subject Average/Minimum/Maximum/Std Dev; turn off totals	Four rows; Maths 80.20 / 58 / 94 / 12.75.

Place the four visuals on one report page titled Students' Exam Scores. This is the requested desktop dashboard layout. Capture imported data types, the DAX measure list/formula bar and the completed report page. A report page can be prepared locally without publishing to the Power BI service.

Q4 | Python method

Aim: read the CSV with pandas, inspect the data, compute statistics and performance measures, then produce the required plots. (10 marks)

Method

a. Read the CSV; print the first five rows, column names and data types.
b. For each subject, compute mean, median, statistical mode, minimum, maximum, range, sample variance and sample standard deviation.
c. Add Total and Percentage; identify all rows tied for the maximum/minimum total.
d. Plot Maths against the categorical student IDs and make a histogram.
e. Plot the four subject means.
f. Plot total marks by Student_ID and print adjacent-ID changes.

Sample statistics use n − 1 in the denominator. A frequency table prevents a unique observation from being misleadingly presented as a mode. The loader checks column names, missing values, duplicate IDs and score bounds.

```
python -m pip install -r requirements.txt
python students_analysis.py
python -m unittest -v
```

The program saves tables and PNG charts under outputs/python. Python 3.12.14, pandas 2.2.3 and matplotlib 3.10.8 were used for the recorded run.

Q4 | Python | code 1/3

File: students_analysis.py. Full, selectable source; line continuations are indented.

```
"""MCSL-065 Q4: inspect, summarise and plot the supplied exam scores."""

import argparse
from pathlib import Path

import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
import pandas as pd

SUBJECTS = ["Physics", "Chemistry", "Maths", "English"]
COLORS = ["#26756c", "#d38b38", "#476f9d", "#a25777"]


def load_scores(path):
    frame = pd.read_csv(path)
    required = ["Student_ID", "Student_Name", *SUBJECTS]
    if list(frame.columns) != required:
        raise ValueError("CSV columns do not match the assignment dataset.")
    if frame.empty or frame.isna().any().any():
        raise ValueError("The dataset must be nonempty and contain no missing values.")
    if frame.Student_ID.duplicated().any():
        raise ValueError("Student_ID must be unique.")
    for subject in SUBJECTS:
        frame[subject] = pd.to_numeric(frame[subject], errors="raise")
        if not frame[subject].between(0, 100).all():
            raise ValueError("Each subject score must be between 0 and 100.")
    return frame


def subject_statistics(frame):
    rows = []
    for subject in SUBJECTS:
        values = frame[subject]
        counts = values.value_counts()
        modes = sorted(counts[counts == counts.max()].index)
        mode = "No mode" if counts.max() == 1 else ", ".join(map(str, modes))
        rows.append({
            "Subject": subject, "Mean": values.mean(),
            "Median": values.median(), "Mode": mode,
            "Minimum": values.min(), "Maximum": values.max(),
            "Range": values.max() - values.min(),
            "Variance": values.var(ddof=1), "Std_Dev": values.std(ddof=1)
        })
    return pd.DataFrame(rows).set_index("Subject")


def make_plots(frame, output):
    plt.rcParams.update({"font.family": "DejaVu Sans", "font.size": 11})
    def save(name):
        plt.tight_layout()
```

Q4 | Python | code 2/3

File: students_analysis.py. Full, selectable source; line continuations are indented.

```
        plt.savefig(output / name, dpi=180, bbox_inches="tight")
        plt.close()
    plt.figure(figsize=(8, 4.2))
    plt.scatter(frame.Student_ID, frame.Maths, s=65, color=COLORS[2])
    plt.axhline(frame.Maths.mean(), color=COLORS[0], linestyle="--",
                label=f"Mean = {frame.Maths.mean():.1f}")
    plt.xlabel("Student ID (category)"); plt.ylabel("Maths marks / 100")
    plt.title("Maths scores by student"); plt.ylim(50, 100); plt.legend()
    save("python_maths_scatter.png")
    plt.figure(figsize=(8, 4.2))
    counts, _, bars = plt.hist(frame.Maths, bins=[50, 60, 70, 80, 90, 100],
                               color=COLORS[2], edgecolor="white")
    for count, bar in zip(counts, bars):
        plt.text(bar.get_x()+bar.get_width()/2, count+0.06, str(int(count)), ha="center")
    plt.xticks([50, 60, 70, 80, 90, 100]); plt.yticks(range(5)); plt.ylim(0, 4)
    plt.xlabel("Maths marks (10-mark intervals)"); plt.ylabel("Number of students")
    plt.title("Distribution of Maths scores")
    save("python_maths_histogram.png")
    plt.figure(figsize=(8, 4.2))
    means = frame[SUBJECTS].mean()
    bars = plt.bar(SUBJECTS, means, color=COLORS, width=0.58)
    plt.bar_label(bars, fmt="%.2f", padding=4); plt.ylim(0, 100)
    plt.xlabel("Subject"); plt.ylabel("Average marks / 100")
    plt.title("Subject-wise average scores")
    save("python_subject_averages.png")
    plt.figure(figsize=(8, 4.2))
    plt.plot(frame.Student_ID, frame.Total, marker="o", color=COLORS[0])
    plt.axhline(frame.Total.mean(), color="#6b7280", linestyle="--",
                label=f"Mean total = {frame.Total.mean():.1f}")
    plt.xlabel("Student ID (category)"); plt.ylabel("Total marks / 400")
    plt.title("Total marks across students"); plt.ylim(200, 400); plt.legend()
    save("python_total_trend.png")


def main():
    parser = argparse.ArgumentParser(description=__doc__)
    parser.add_argument("--csv", type=Path,
default=Path(__file__).with_name("students_scores.csv"))
    parser.add_argument("--output", type=Path, default=Path(__file__).parent /
"outputs/python")
    args = parser.parse_args(); args.output.mkdir(parents=True, exist_ok=True)
    frame = load_scores(args.csv)
    print("A. FIRST FIVE ROWS\n", frame.head().to_string(index=False), sep="")
    print("\nCOLUMN NAMES:", ", ".join(frame.columns))
    print("\nDATA TYPES\n", frame.dtypes.to_string(), sep="")
    summary = subject_statistics(frame)
    print("\nB. DESCRIPTIVE STATISTICS (sample variance and SD)")
    print(summary.round(4).to_string())
    summary.to_csv(args.output / "descriptive_statistics.csv")
    frame["Total"] = frame[SUBJECTS].sum(axis=1)
    frame["Percentage"] = frame.Total / 400 * 100
```

Q4 | Python | code 3/3

File: students_analysis.py. Full, selectable source; line continuations are indented.

```
101      print("\nC. TOTALS AND PERCENTAGES\n", frame.to_string(index=False), sep="")
102      for label, total in [("Topper", frame.Total.max()), ("Lowest scorer",
     frame.Total.min())]:
103          for _, row in frame.loc[frame.Total == total].iterrows():
104              print(f"{label}: {row.Student_Name} ({row.Student_ID}), {row.Total}/400,
     {row.Percentage:.2f}%")
105      frame.to_csv(args.output / "scores_with_totals.csv", index=False)
106      make_plots(frame, args.output)
107      print("\nD-F. Saved scatter, histogram, subject-average and total-trend plots.")
108      print("Maths spans 58-94; 6 of 10 students score at least 80.")
109      print("IDs identify different students; the connected totals are not a time series.")
110      changes = frame.Total.diff()
111      for i in range(1, len(frame)):
112          print(f"{frame.Student_ID.iloc[i-1]} to {frame.Student_ID.iloc[i]}:
     {changes.iloc[i]:+g}")
113
114
115  if __name__ == "__main__":
116      main()
```

Q4 | Python inspection and statistics

Python: inspection and statistics

```
python students_analysis.py
```

```
A. FIRST FIVE ROWS
Student_ID Student_Name  Physics  Chemistry  Maths  English
       S01        Rahul       78         72     85       80
       S02       Ananya       88         91     90       86
       S03         Amit       65         69     70       72
       S04         Neha       92         89     94       90
       S05        Karan       55         60     58       62

COLUMN NAMES: Student_ID, Student_Name, Physics, Chemistry, Maths, English

DATA TYPES
Student_ID      object
Student_Name    object
Physics          int64
Chemistry        int64
Maths            int64
English          int64

B. DESCRIPTIVE STATISTICS (sample variance and SD)
            Mean  Median     Mode  Minimum  Maximum  Range  Variance  Std_Dev
Subject
Physics     76.4    79.5  No mode       55       92     37  173.1556  13.1589
Chemistry   77.9    78.5  No mode       60       92     32  149.4333  12.2243
Maths       80.2    86.5       90       58       94     36  162.6222  12.7523
English     79.2    82.0  No mode       62       91     29  102.4000  10.1193
```

Actual execution transcript · Exit status: 0

Figure 4. Captured output of the executed Python program: head(), column names, data types and all eight requested statistics.

Q4 | Totals and performance differences

Python: totals and performance

```
python students_analysis.py
```

```
C. TOTALS AND PERCENTAGES
Student_ID Student_Name  Physics  Chemistry  Maths  English  Total  Percentage
       S01        Rahul       78         72     85       80    315       78.75
       S02       Ananya       88         91     90       86    355       88.75
       S03         Amit       65         69     70       72    276       69.00
       S04         Neha       92         89     94       90    365       91.25
       S05        Karan       55         60     58       62    235       58.75
       S06        Priya       81         85     88       84    338       84.50
       S07        Rohit       70         68     75       73    286       71.50
       S08       Simran       90         92     89       91    362       90.50
       S09        Arjun       60         65     63       67    255       63.75
       S10        Pooja       85         88     90       87    350       87.50
Topper: Neha (S04), 365/400, 91.25%
Lowest scorer: Karan (S05), 235/400, 58.75%

D-F. Saved scatter, histogram, subject-average and total-trend plots.
Maths spans 58-94; 6 of 10 students score at least 80.
IDs identify different students; the connected totals are not a time series.
S01 to S02: +40
S02 to S03: -79
S03 to S04: +89
S04 to S05: -130
S05 to S06: +103
S06 to S07: -52
S07 to S08: +76
S08 to S09: -107
S09 to S10: +95
```

Actual execution transcript · Exit status: 0

Figure 5. Actual Python totals, percentages, highest/lowest scorers and adjacent-student differences.

The largest decrease between adjacent IDs is S04 → S05 (365 → 235, −130). The largest increase is S05 → S06 (235 → 338, +103). These are differences between people, not changes in one student’s performance over time.

Q4 | Maths scatter and histogram

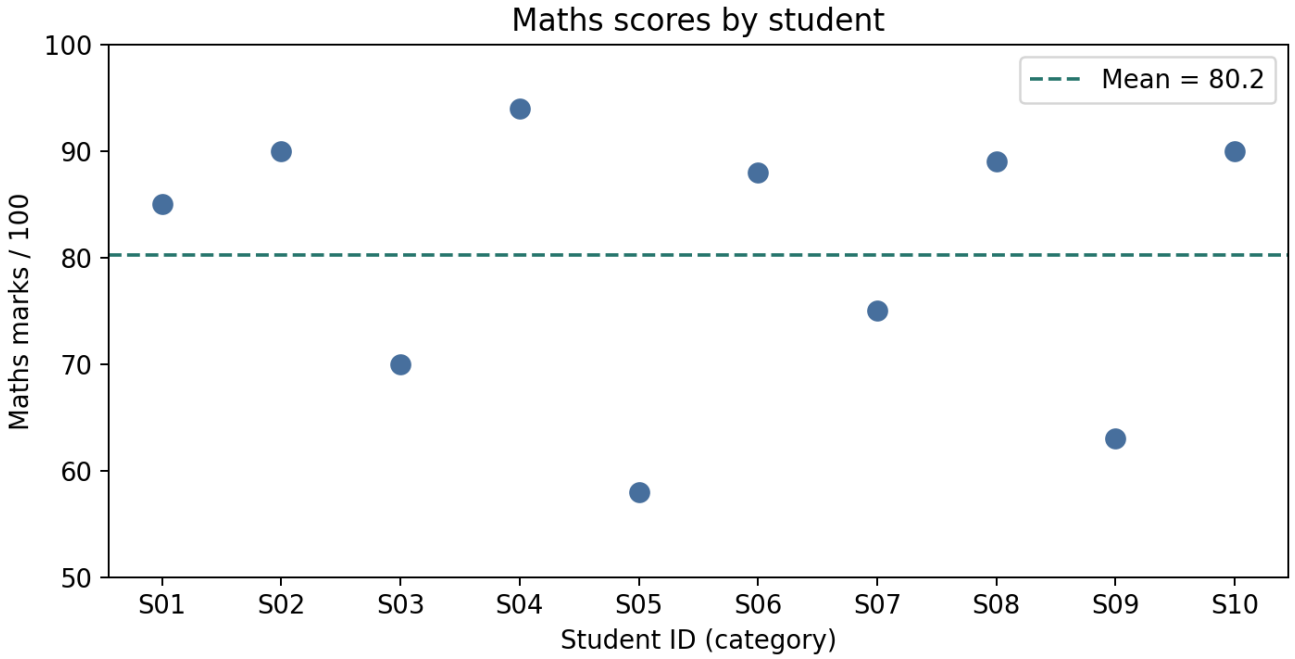


Figure 6. Python-generated scatter diagram. Student ID is used only as a categorical position.

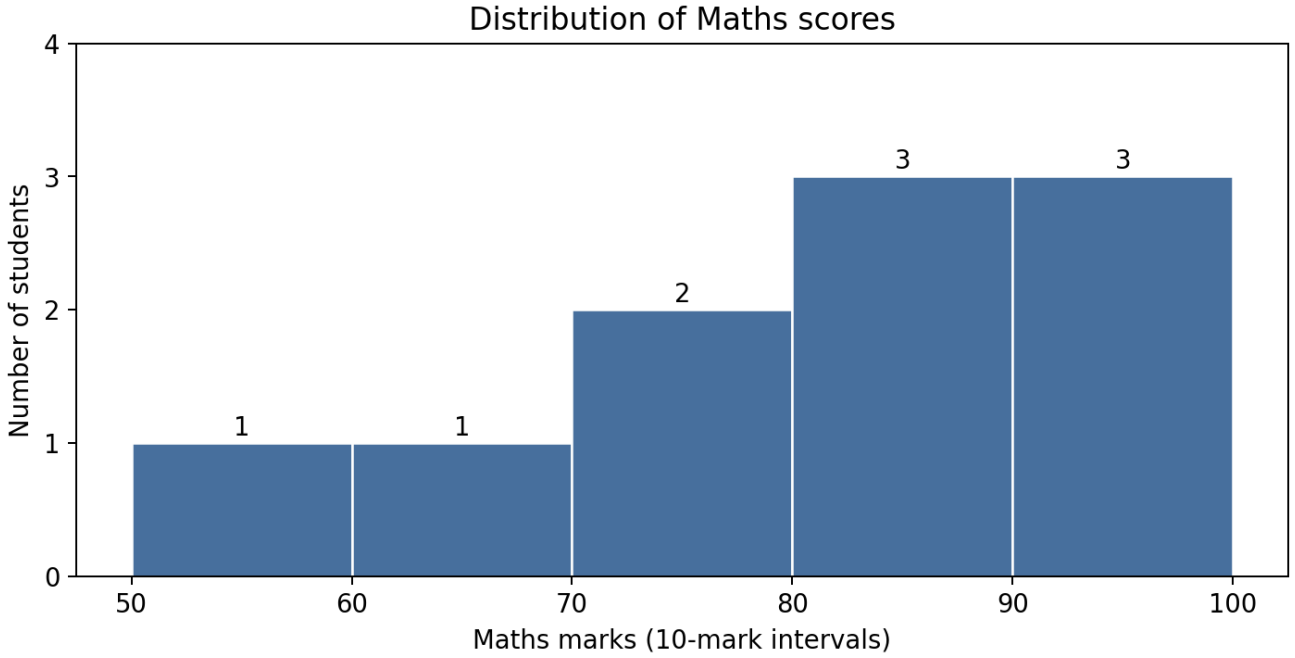


Figure 7. Python-generated histogram with frequencies 1, 1, 2, 3, 3.

Maths scores range from 58 to 94 (range 36), with sample SD 12.75. Six of ten scores are at least 80, so the larger concentration is in the upper two intervals, but the full set still has appreciable spread.

Q4 | Subject means and total marks

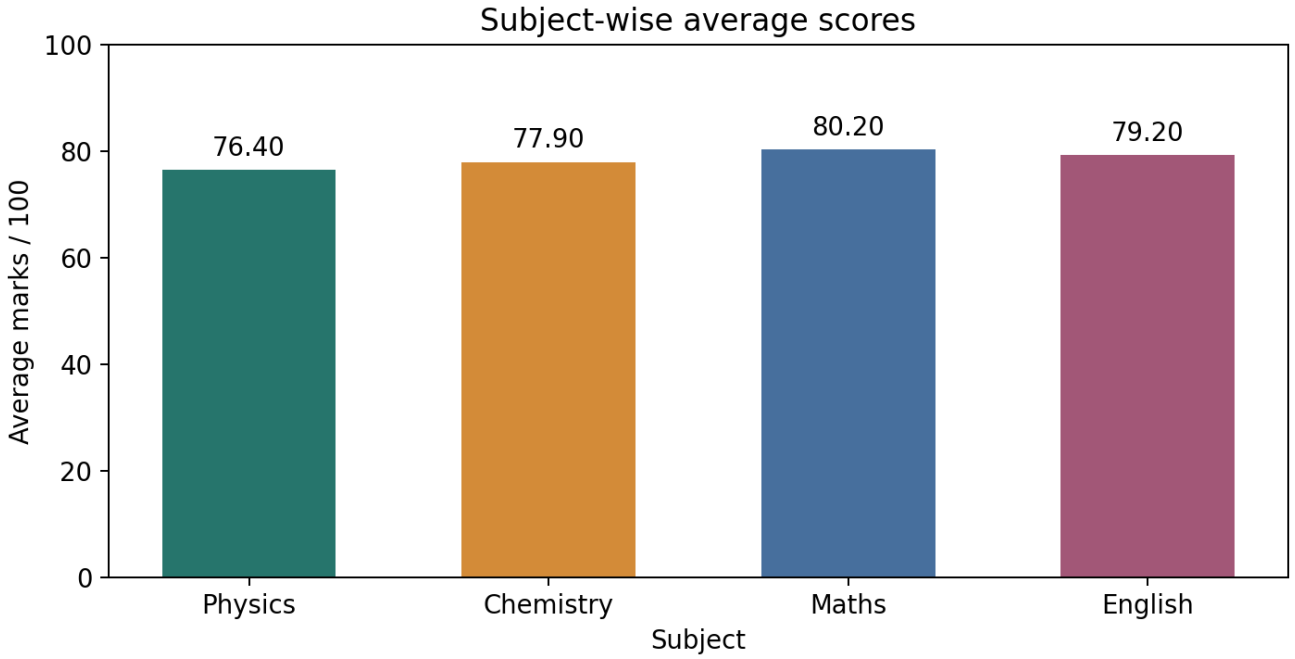


Figure 8. Python-generated bar chart of subject averages.

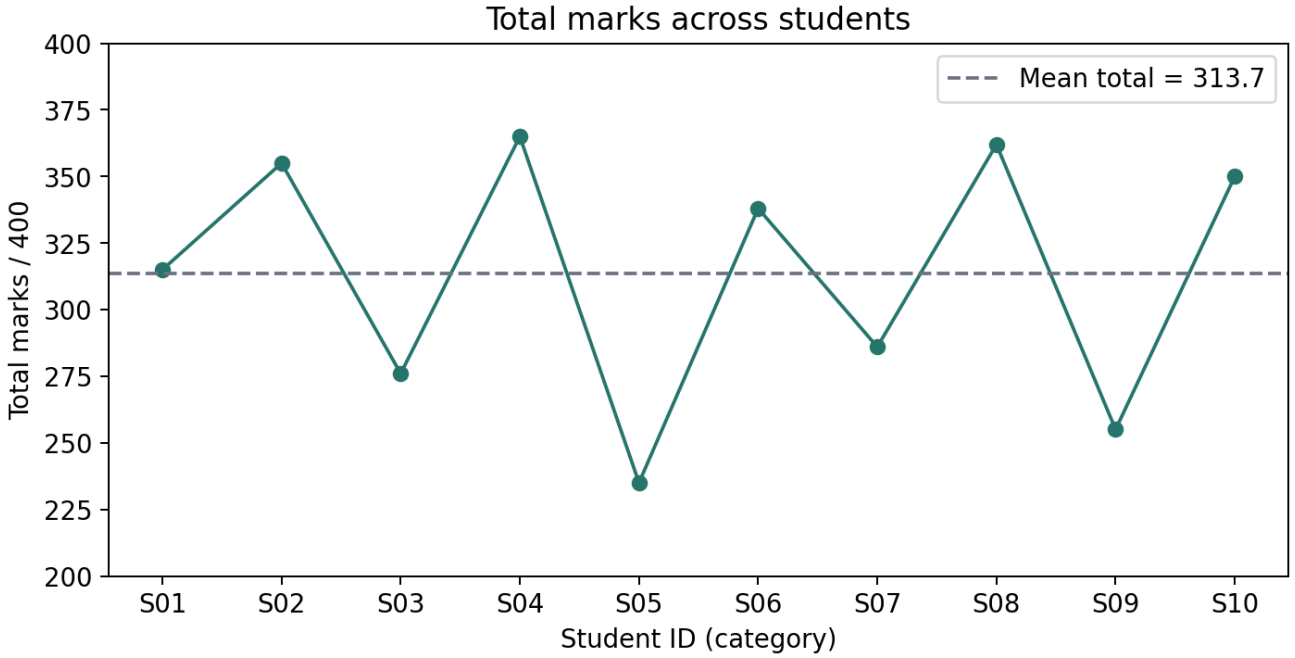


Figure 9. Python-generated line chart of total marks in Student_ID order.

Mean subject marks are close, spanning 76.40–80.20. Total marks are much more varied, spanning 235–365. Neha and Simran have the two largest totals, while Karan and Arjun have the two smallest.

Q5 | R method

Aim: perform the same analysis using a base R data frame and graphics. (10 marks)

read.csv() loads the records; head(), str() and summary() inspect them. The statistical_mode() function uses frequencies, because R’s mode() describes an object’s storage type rather than its statistical mode. var() and sd() already use the sample denominator n − 1.

rowSums() computes total marks and Percentage = Total / 400 × 100. Logical conditions retain all tied highest/lowest scorers. The script saves two PNGs containing the four required plots and exports the same summary tables as Python.

```
Rscript students_analysis.R
# Optional input/output arguments:
Rscript students_analysis.R students_scores.csv outputs/r
```

The recorded run used an actual R 4.5.0 interpreter with base packages. Python and R numeric summaries, modes, totals and percentages were compared and agreed to numerical tolerance.

Q5 | R | code 1/2

File: students_analysis.R. Full, selectable source; line continuations are indented.

```r
# MCSL-065 Q5: base R solution. Run: Rscript students_analysis.R
args <- commandArgs(trailingOnly = TRUE)
csv_path <- if (length(args) >= 1) args[1] else "students_scores.csv"
out <- if (length(args) >= 2) args[2] else "outputs/r"
dir.create(out, recursive = TRUE, showWarnings = FALSE)
students <- read.csv(csv_path, stringsAsFactors = FALSE)
subjects <- c("Physics", "Chemistry", "Maths", "English")
stopifnot(nrow(students) > 0, !anyNA(students),
          !anyDuplicated(students$Student_ID),
          all(subjects %in% names(students)))
stopifnot(all(vapply(students[subjects], is.numeric, logical(1))))
stopifnot(all(as.matrix(students[subjects]) >= 0),
          all(as.matrix(students[subjects]) <= 100))

cat("A. FIRST FIVE ROWS\n"); print(head(students, 5), row.names = FALSE)
cat("\nSTRUCTURE\n"); str(students)
cat("\nSUMMARY OF THE DATASET\n"); print(summary(students))

statistical_mode <- function(x) {
  counts <- table(x)
  if (max(counts) == 1) return("No mode")
  paste(names(counts)[counts == max(counts)], collapse = ", ")
}
statistics <- do.call(rbind, lapply(subjects, function(subject) {
  x <- students[[subject]]
  data.frame(Subject = subject, Mean = mean(x), Median = median(x),
             Mode = statistical_mode(x), Minimum = min(x), Maximum = max(x),
             Range = max(x) - min(x), Variance = var(x), Std_Dev = sd(x))
}))
cat("\nB. DESCRIPTIVE STATISTICS (sample variance and SD)\n")
print(statistics, row.names = FALSE, digits = 6)
write.csv(statistics, file.path(out, "descriptive_statistics.csv"), row.names = FALSE)

students$Total <- rowSums(students[subjects])
students$Percentage <- students$Total / 400 * 100
cat("\nC. TOTALS AND PERCENTAGES\n"); print(students, row.names = FALSE)
cat("\nTOPPER\n")
print(students[students$Total == max(students$Total), ], row.names = FALSE)
cat("\nLOWEST SCORER\n")
print(students[students$Total == min(students$Total), ], row.names = FALSE)
write.csv(students, file.path(out, "scores_with_totals.csv"), row.names = FALSE)

colours <- c("#26756c", "#d38b38", "#476f9d", "#a25777")
png(file.path(out, "r_maths_distribution.png"), width = 1600, height = 650, res = 150)
par(mfrow = c(1, 2), mar = c(5, 4, 3, 1))
plot(seq_len(nrow(students)), students$Maths, pch = 19, col = colours[3],
     xaxt = "n", xlab = "Student ID (category)", ylab = "Maths marks / 100",
     main = "Maths scores by student", ylim = c(50, 100))
axis(1, at = seq_len(nrow(students)), labels = students$Student_ID, las = 2)
abline(h = mean(students$Maths), lty = 2, col = colours[1])
hist(students$Maths, breaks = seq(50, 100, 10), right = FALSE,
```

Q5 | R | code 2/2

File: students_analysis.R. Full, selectable source; line continuations are indented.

```
52       col = colours[3], border = "white", labels = TRUE,
53       xlab = "Maths marks (10-mark intervals)", ylab = "Number of students",
54       main = "Distribution of Maths scores", ylim = c(0, 4))
55  invisible(dev.off())
56
57  png(file.path(out, "r_averages_and_totals.png"), width = 1600, height = 650, res = 150)
58  par(mfrow = c(1, 2), mar = c(5, 4, 3, 1))
59  averages <- colMeans(students[subjects])
60  positions <- barplot(averages, col = colours, ylim = c(0, 100),
61                       ylab = "Average marks / 100", main = "Subject-wise average scores")
62  text(positions, averages + 4, sprintf("%.2f", averages))
63  plot(seq_len(nrow(students)), students$Total, type = "b", pch = 19,
64       col = colours[1], xaxt = "n", xlab = "Student ID (category)",
65       ylab = "Total marks / 400", main = "Total marks across students", ylim = c(200, 400))
66  axis(1, at = seq_len(nrow(students)), labels = students$Student_ID, las = 2)
67  abline(h = mean(students$Total), lty = 2, col = "grey50")
68  invisible(dev.off())
69  cat("\nD-F. Plots saved. Maths spans 58-94; 6 of 10 students score at least 80.\n")
70  cat("Student IDs are categories; changes across IDs are not changes over time.\n")
71  cat("Changes in totals between adjacent IDs:\n"); print(diff(students$Total))
```

Q5 | R inspection and summary

R: data frame and structure

Rscript students_analysis.R

```
A. FIRST FIVE ROWS
 Student_ID Student_Name Physics Chemistry Maths English
        S01        Rahul      78        72    85      80
        S02       Ananya      88        91    90      86
        S03         Amit      65        69    70      72
        S04         Neha      92        89    94      90
        S05        Karan      55        60    58      62

STRUCTURE
'data.frame':   10 obs. of  6 variables:
 $ Student_ID  : chr  "S01" "S02" "S03" "S04" ...
 $ Student_Name: chr  "Rahul" "Ananya" "Amit" "Neha" ...
 $ Physics     : int  78 88 65 92 55 81 70 90 60 85
 $ Chemistry   : int  72 91 69 89 60 85 68 92 65 88
 $ Maths       : int  85 90 70 94 58 88 75 89 63 90
 $ English     : int  80 86 72 90 62 84 73 91 67 87
```

Actual execution transcript · Exit status: 0

Figure 10. Actual R output: first five records and data-frame structure.

R: descriptive statistics

Rscript students_analysis.R

```
SUMMARY OF THE DATASET
  Student_ID        Student_Name          Physics        Chemistry
 Length:10          Length:10          Min.   :55.00   Min.   :60.00
 Class :character   Class :character   1st Qu.:66.25   1st Qu.:68.25
 Mode  :character   Mode  :character   Median :79.50   Median :78.50
                                       Mean   :76.40   Mean   :77.90
                                       3rd Qu.:87.25   3rd Qu.:88.75
                                       Max.   :92.00   Max.   :92.00
    Maths          English
 Min.   :58.00   Min.   :62.00
 1st Qu.:71.25   1st Qu.:72.25
 Median :86.50   Median :82.00
 Mean   :80.20   Mean   :79.20
 3rd Qu.:89.75   3rd Qu.:86.75
 Max.   :94.00   Max.   :91.00

B. DESCRIPTIVE STATISTICS (sample variance and SD)
   Subject Mean Median    Mode Minimum Maximum Range Variance Std_Dev
   Physics 76.4   79.5 No mode      55      92    37  173.156 13.1589
 Chemistry 77.9   78.5 No mode      60      92    32  149.433 12.2243
     Maths 80.2   86.5      90      58      94    36  162.622 12.7523
   English 79.2   82.0 No mode      62      91    29  102.400 10.1193
```

Actual execution transcript · Exit status: 0

Figure 11. Actual R summary and subject statistics, including the statistical modes.

Q5 | R totals and scorers

R: totals and performance

```
Rscript students_analysis.R
```

```
C. TOTALS AND PERCENTAGES
 Student_ID Student_Name Physics Chemistry Maths English Total Percentage
        S01         Rahul      78        72    85      80   315      78.75
        S02        Ananya      88        91    90      86   355      88.75
        S03          Amit      65        69    70      72   276      69.00
        S04          Neha      92        89    94      90   365      91.25
        S05         Karan      55        60    58      62   235      58.75
        S06         Priya      81        85    88      84   338      84.50
        S07         Rohit      70        68    75      73   286      71.50
        S08        Simran      90        92    89      91   362      90.50
        S09         Arjun      60        65    63      67   255      63.75
        S10         Pooja      85        88    90      87   350      87.50

TOPPER
 Student_ID Student_Name Physics Chemistry Maths English Total Percentage
        S04          Neha      92        89    94      90   365      91.25

LOWEST SCORER
 Student_ID Student_Name Physics Chemistry Maths English Total Percentage
        S05         Karan      55        60    58      62   235      58.75

D-F. Plots saved. Maths spans 58-94; 6 of 10 students score at least 80.
Student IDs are categories; changes across IDs are not changes over time.
Changes in totals between adjacent IDs:
[1]   40  -79   89 -130  103  -52   76 -107   95
```

Actual execution transcript · Exit status: 0

Figure 12. Actual R output of totals, percentages, topper, lowest scorer and adjacent-ID differences.

Q5 | R graphics and interpretation

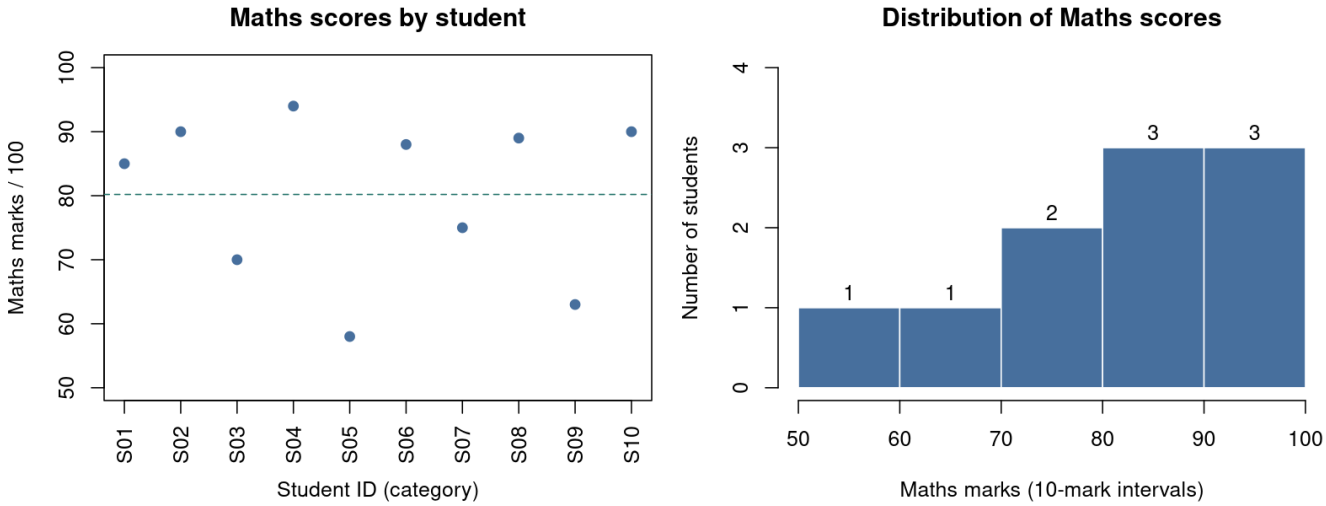


Figure 13. R-generated Maths scatter diagram and histogram.

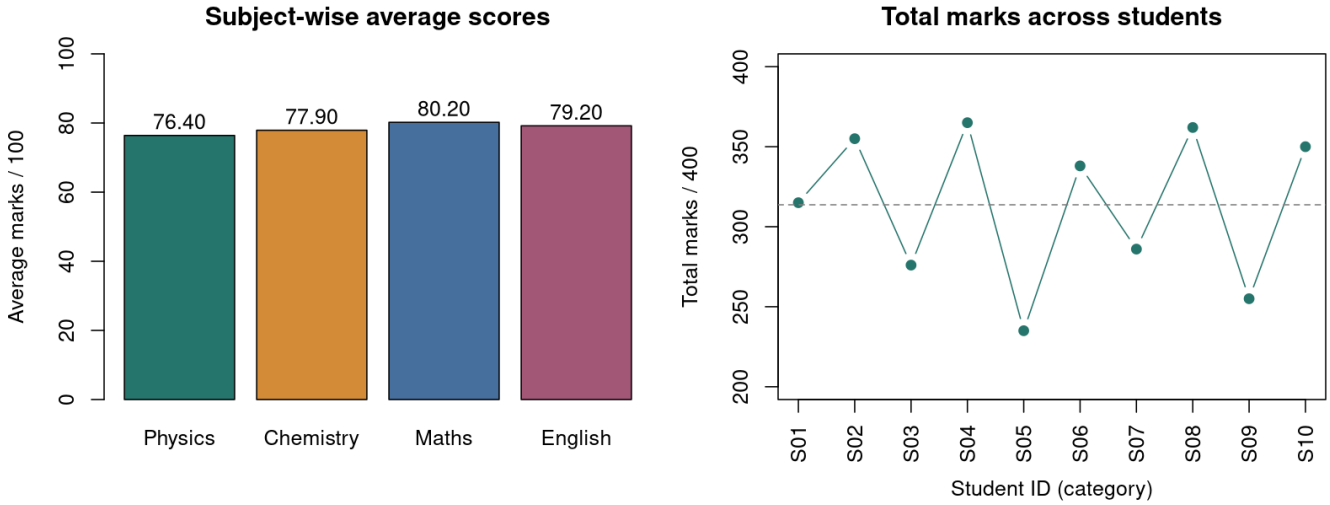


Figure 14. R-generated subject-average bars and total-mark line chart.

R confirms the Python findings: Maths has mean 80.20, range 36 and sample SD 12.75; six students score at least 80. The class mean total is 313.70. The line rises and falls because it joins different students; the order is not a time variable.

Checks and viva preparation

Check	Recorded outcome
Source data	Ten records transcribed from the question paper; totals verified.
Python tests	4 tests passed: fixed results, modes, multiple modes, and invalid input.
Python ↔ R	All numeric summaries, modes, totals and percentages agree.
Excel formula validation	122 formulas recalculated in LibreOffice; no mismatches.
Native GUI evidence	Excel ToolPak, Tableau and Power BI application screenshots remain pending.

Viva question	Answer
Why use sample variance?	It estimates variance using the $n - 1$ denominator. Keep the same convention in every tool.
Does R mode() find the most common value?	No. It reports storage mode; use table() and compare frequencies.
What is the meaning of $p > 0.05$?	The test does not reject its null at the chosen threshold; this does not prove equality.
What does a correlation near +1 mean?	A strong positive linear association in the observed data, not causation.
Why are subject pie shares near 25%?	The four subject averages are close, so each contributes about a quarter of their sum.
What is a Power BI measure?	A DAX calculation evaluated in the visual’s current filter context.
How is the Tableau Maths filter scoped?	Apply it only to the Maths worksheet, leaving all four subjects in other views.

Sources and final submission checks

Primary source: MSCDSA-MCSL-065-2026.pdf supplied with this task. It defines the dataset, five tool tasks and their marks. The supplied MCSL-065 reference solution was reviewed for coverage; its template and dashboard blueprint were not reused.

Official programme-guide check

An HTTPS request to IGNOU's official programme page and assignment service was denied by the environment's network proxy (403). Domain additions were saved to the environment configuration draft, but this does not itself apply the network change. The official programme guide has therefore not been verified online in this review copy.

The supplied papers state that assignment submission and the required lab attendance are prerequisites for the practical examination, and that assignment viva voce is compulsory. Confirm the current programme-guide and Study Centre instructions before final submission. Do not treat this review copy as certification of institutional formatting compliance.

Technical documentation for further checking

pandas: pandas.pydata.org/docs/
R: cran.r-project.org/manuals.html
Excel Analysis ToolPak: support.microsoft.com (Analysis ToolPak documentation)
Power BI DAX: learn.microsoft.com/dax/
Tableau Desktop: help.tableau.com/current/pro/desktop/en-us/

Before submitting

1. Fill the Study Centre field and actual submission date; sign the cover.
2. Run the Excel Analysis ToolPak and capture its results and charts.
3. Build and capture the Tableau dashboard.
4. Validate the DAX measures and capture the Power BI report page.
5. Insert native screenshots in the editable DOCX, remove the pending-evidence notices, and export the final PDF.
6. Review the official programme guide and prepare to explain the code and results in the compulsory viva.